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A Coupled Chemo-Mechanical and Life-Cycle Framework for Oyster-Shell-Derived Cementitious Materials in Aggressive Infrastructure Environments

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Abstract

Oyster-shell powder is promoted as a low-carbon cement substitute, but joint carbon and durability benefits are usually asserted rather than tested. This paper asks under which processing, substitution and exposure conditions shell-derived concrete delivers a joint advantage over ordinary Portland cement (OPC) concrete, and where that claim fails. Two pathways are separated: uncalcined shell powder replacing cement (P1) and shell calcium carbonate replacing limestone in the kiln feed (P2). A coupled model links hydration, porosity, chloride, acid and leaching transport, damage accumulation (fatigue, freeze–thaw, shrinkage, thermal, abrasion) and cradle-to-gate carbon, and is propagated through paired Latin hypercube Monte Carlo with Sobol sensitivity and an explicit falsification region for a pavement, a dam and a marine culvert. With inert shell filler the model over-predicts the strength lost to replacement; it was therefore tested out of sample against published scallop-shell concrete data (Abbas and Kudukkan 2025), a filler-efficiency parameter was calibrated by rejection approximate Bayesian computation on two mixes, and a third mix was held out. Calibration changes the conclusions. At r = 25% and constant water-to-binder ratio the calibrated median carbon change is −21% with strength 0.91 of OPC, and the probability of a carbon cut of at least 5% with non-inferior life rises from 0.01 to 0.23 in the pavement case but remains 0.00 in the marine culvert, where chloride-driven life is shorter in almost every draw. The P2 pathway raises carbon by about 5.1% unless calcination CO_2 is credited as biogenic. Chloride transport in filler-bearing paste is not yet calibrated, so all results are conditional on stated assumptions, and the experiments needed to test them are specified.

Keywords: oyster shell; calcium carbonate filler; service-life prediction; embodied carbon; approximate Bayesian calibration; uncertainty and sensitivity analysis; falsification

1. Introduction

Cement production is a major source of anthropogenic CO_2, with roughly half of the process emissions arising from the calcination of limestone rather than from fuel (Andrew 2019). Strategies to reduce this footprint include clinker substitution by supplementary materials and fillers, alternative binders and carbon capture (Scrivener et al. 2018). Aquaculture waste shells, composed mainly of calcium carbonate, are an attractive candidate because they are a locally available, otherwise problematic waste. Technical communications on “oyster-shell green concrete” frequently claim large carbon reductions, fire resistance to very high temperatures, accelerated hydration, reduced shrinkage and superior chemical resistance. These

claims are presented without a mathematical model, uncertainty analysis or experimental evidence, and they conflate materials pathways that are chemically and mechanically different.

1.1 What existing research establishes

Shell-derived calcium carbonate. Ground shells are predominantly calcium carbonate with a small organic matrix and trace salts. Experimental studies of ground seashells (including oyster shell) as partial cement replacement in mortars and masonry blocks report that compressive strength decreases as the replacement fraction increases, that water absorption increases in blocks, that setting is retarded, and that low replacement levels (about 10%) can still meet standards, with higher levels reaching required strength only at later ages (Lertwattanaruk et al. 2012; Jasni et al. 2024). Drying shrinkage of mortars containing some shell types was reported to decrease (Lertwattanaruk et al. 2012). These studies do not provide the kinetic, transport and damage parameters needed for a predictive treatment, and they do not address carbon and life together. A recent constant-binder study of scallop-shell powder reports about 25% lower 28-day strength at 40% binder replacement and a lower carbon footprint (Abbas and Kudukkan 2025); it is the out-of-sample dataset used in Section 4.4.

Hydration and the filler effect. Fine calcium carbonate accelerates early cement hydration through heterogeneous nucleation of C–S–H and, in the presence of aluminates, forms carboaluminates that stabilise ettringite (Lothenbach et al. 2008; Berodier and Scrivener 2014). Dilution, however, raises the effective water-to-cement ratio and removes reactive material, so strength generally decreases with replacement. Organic matter associated with shells may retard hydration.

Transport and durability. Chloride ingress is commonly modelled by Fick's second law with an age-dependent diffusivity; the Life-365 formulation relates the 28-day diffusivity of OPC concrete to the water-to-cementitious ratio and uses an age exponent of 0.2 (Thomas and Bentz 2002). Acid attack and calcium leaching are classically described by moving-front models in which front depth grows with the square root of time and decreases with the acid-neutralising or calcium capacity of the matrix. Calcite can neutralise acid but also dissolves, so its net effect depends on exposure.

Damage and service life. Fatigue of plain concrete follows stress–life (S–N) relations such as that of Aas-Jakobsen (1970); freeze–thaw damage is governed by the degree of saturation relative to a critical value (Fagerlund 1977); reinforcement corrosion is treated with an initiation plus propagation framework (Tuutti 1982); and continuum damage mechanics treats degradation through an effective stress $\sigma/(1 - D)$ (Kachanov 1958; Lemaitre 1996).

Life-cycle assessment. Cradle-to-gate embodied carbon is dominated by the cement. Shell powder replaces cement at a small fraction of its footprint, but processing (washing, drying, grinding) and transport are non-zero, and for P2 the calcination CO_2 per unit $CaCO_3$ is stoichiometrically independent of the carbonate source.

A competing use of shell. Shell is also a restoration resource. It is a widely used substrate for oyster-reef restoration, its supply can limit restoration (Brumbaugh and Coen 2009; Petrolia and Mohrman 2025), and reefs provide habitat, filtration and shoreline services (Beck et al. 2011; Zu Ermgassen et al. 2012; Grabowski et al. 2012). Shell used in concrete is therefore not free of an opportunity cost. That trade-off is the subject of a companion paper on the allocation of recovered shell between reef restoration and construction (Sandler 2026b); here it is labelled pathway P3 and used as context only, and this paper is restricted to the concrete pathways P1 and P2.

1.2 Research gap, claims and hypotheses

A title-level screen of the Crossref index on 1 November 2026 (terms combining oyster or seashell with cement replacement, strength, service life, chloride, life-cycle carbon and uncertainty) returned experimental studies and reviews of shell as a cement replacement but, to the author's knowledge, no model that couples processing-dependent carbon, hydration, chloride and chemical transport, damage accumulation and infrastructure-specific service life under shared parametric uncertainty. This is a limited screen and not a systematic review, so the gap is stated conditionally. The source technical white paper made seven performance claims. Rather than accept them, each is converted into a testable hypothesis (Table 1).

Table 1. Claims in the source white paper and how each is treated in this manuscript.

Claim	Treatment here
Up to 80% CO_2 reduction	Cradle-to-gate carbon model with explicit processing, transport, calcination and credit terms; carbon change reported at constant w/b and at constant strength
Thermal resistance to 2,400 °F	Not modelled; assessed against calcite thermochemistry only (Section 8)
100% limestone substitution	Separated: P2 (feedstock replacement) vs P1 (cement replacement); never placed on one axis
Accelerated hydration, reduced shrinkage	Hydration kinetics with explicit filler acceleration and organic retardation; shrinkage and early-age cracking indices
Superior acid and chemical resistance	Moving-front acid and leaching models in which calcite both buffers and dilutes
Higher structural durability	Service life from damage accumulation and chloride corrosion limit states, paired with an OPC reference
Inorganic-polymer (alkali-activated) bonding	Excluded (pathway P4); shell calcium carbonate is not an aluminosilicate precursor

Research question. Under which processing, substitution and exposure conditions does oyster-shell-derived concrete provide a simultaneous carbon and durability advantage over conventional concrete, and under which does it fail? Formally, we define a falsification region (Section 3.5) and estimate its probability under parameter uncertainty. The hypotheses are: H1, shell replacement at constant w/b lowers carbon but not service life; H2, at constant strength the carbon benefit disappears; H3, the P2 pathway offers no carbon advantage unless calcination CO_2 is credited; H4, shell calcite improves acid resistance only when its particles are fine enough to be accessible.

2. Materials and processing pathways

Oyster shell is approximately 95–98% calcium carbonate (predominantly calcite in the foliated layers) with 1–4 wt% organic matrix (conchiolin) and trace sodium, magnesium, chloride, phosphorus and sulfur; these ranges are generic and should be replaced by measurements for a given supply. Processing for cementitious use comprises collection, weathering or washing to reduce organics and salt, drying, crushing and fine grinding to a target median particle size (10–200 μm is explored). Four pathways are distinguished (Table 2).

Table 2. Pathways considered. They are not interchangeable and are modelled on separate axes.

Pathway	Material route	Principal mechanisms	Status here
P1	Uncalcined shell powder replaces a mass fraction r of cement in concrete	Dilution, filler nucleation, packing, organic retardation, calcite acid buffering	Modelled; r = 10–100%, median size 10–200 µm
P2	Shell $CaCO_3$ replaces a fraction ρ of limestone in clinker raw meal; cement chemistry unchanged	Impurity effects on cement; processing and transport carbon; calcination CO_2 unchanged per kg $CaCO_3$	Modelled; ρ = 25–100%, biogenic credit 0–100%
P3	Recovered, cleaned shell returned to the water as reef substrate (competing use)	Recruitment substrate; habitat, filtration and shoreline services; transport and cleaning energy	Context only; allocation modelled in the companion paper (Sandler 2026b)
P4	Shell-based alkali-activated binders	Requires an aluminosilicate precursor; not a calcium-carbonate substitution	Out of scope

3. Theoretical framework

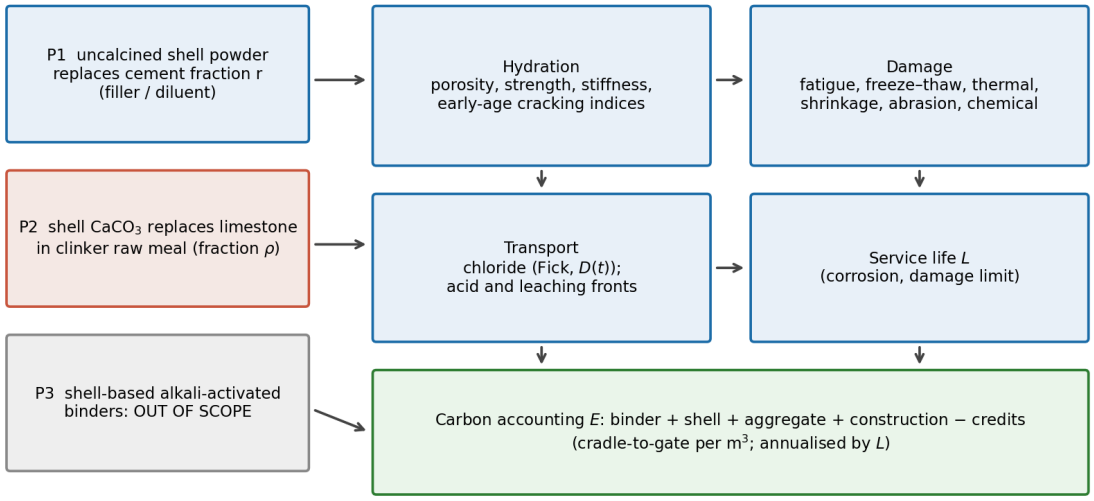


Fig. 1. Coupled model architecture. Pathways P1 and P2 feed hydration, which sets porosity, strength and stiffness; these drive transport and damage, which determine service life; carbon accounting is evaluated per m^3 and per year of life.

All quantities are in SI-consistent units; Appendix B lists symbols and units. Damage D, stress ratio S and degree of saturation are dimensionless, and every rate in the damage law has units of inverse time. All parameters are uncertain and are sampled (Section 4). This is a **proposed phenomenological framework**, not a validated constitutive model; each component is anchored to an established relation where one exists and its limits are stated.

3.1 Hydration, porosity and strength

Hydration degree α evolves with a reaction-progress law that can be accelerated or retarded by shell addition:

$$d\alpha/dt = k_0\, A(T)\, (1 + \nu\, s(d)\, r)\, f_{ret}\, (\alpha_{max} - \alpha)^n \quad (1)$$

where k_0 is the rate constant at 20 °C, $A(T) = \exp[E_a/R\,(1/293.15 - 1/T)]$ is the Arrhenius factor, ν is the filler nucleation coefficient, $s(d) = \min[(30\ \mu m/d)^{0.5}, 2]$ is a fineness factor, $f_{ret} = 1/(1 + \kappa_o\, q)$ is the retardation factor with $q = o(1 - \eta_w)\, r/(1 - r)$ the residual organic dose per unit cement (o is the organic

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fraction and η_w the washing efficiency), and α_max = min[α_cap, (w/c)/0.42] limits the attainable degree to the water available per unit cement. For α(0) = 0 and n ≠ 1 the closed form is

$$\alpha(t) = \alpha_{max} - [\alpha_{max}^{(1-n)} + (n-1)\, k\, t]^{(1/(1-n))} \quad (2)$$

Shell addition therefore raises the rate through ν and the ceiling through a larger effective w/c, but lowers the rate through organics, and removes cement. With cement volume v_c, water volume v_w, hydration products of volume 2.06 α v_c, the capillary porosity is φ_cap = v_w − 1.06 α v_c (Powers and Brownyard 1947) and the gel/space ratio X = 2.06 α v_c/(v_w + α v_c). Compressive strength is

$$f_c = S_0' X^{\kappa} [1 + \beta_p\, s\, r(1-r)] (1 - k_{imp}\, \rho) \quad (3)$$

where the exponent κ is uncertain (1.8–3.0) and $S_0' = S_0 \times 0.742^{(2-\kappa)}$ keeps the nominal OPC strength at w/c = 0.45 fixed for any κ; β_p is a packing bonus and k_imp an impurity penalty that applies only to P2. Stiffness, tensile and flexural strength follow $E = 4700\sqrt{f_c}$, $f_t = 0.33\sqrt{f_c}$ and $f_r = 0.62\sqrt{f_c}$ (MPa; ACI 318).

Filler-efficiency extension. In the original model shell powder is an inert diluent in the strength relation. To test this against published data (Section 4.4) an optional filler efficiency k_f is introduced: the cement volume entering the gel/space ratio is multiplied by [1 + k_f r/(1 − r)], so that a unit mass of shell counts as k_f units of cement for strength, as in the classical k-value concept for fillers and supplementary materials. Hydration, porosity and transport are not changed. The value k_f = 0 recovers the original model; the prior is U(0, 1.2).

3.2 Transport and chemical degradation

Chloride. One-dimensional diffusion with spatially uniform, age-dependent diffusivity has the solution $C(x,t) = C_s\, \mathrm{erfc}[x/(2\sqrt{\tau})]$ with $\tau = \int D\, dt$ (Crank 1975). The diffusivity is

$$D(t) = D_{28}\, f_T\, (28\text{ d} / t_{age})^{m}\, [1 + \kappa_c\, D_{mech}(t)], \quad m = m_0 + \omega\, r \quad (4)$$

where f_T is an Arrhenius temperature factor (activation energy 35–50 kJ/mol; Life-365 uses 44.6 kJ/mol), $m_0 \approx 0.2$ (Thomas and Bentz 2002), ω an uncertain filler shift and κ_c couples mechanical damage to transport (cracks raise diffusivity). Because the damage trajectory does not depend on chloride, the coupling is exact for constant surface chloride. The 28-day diffusivity is a blend of two mappings with an uncertain weight χ_D ~ U(0,1), which represents **model-form uncertainty**:

$$\ln D_{28} = (1 - \chi_D) \ln D_{\varphi} + \chi_D \ln D_{wc} \quad (5)$$

$D_{\varphi} = \lambda_D\, D_{ref}\, (\varphi_p/\varphi_{ref})^{(\zeta + \Delta\zeta)}$ is a porosity mapping, with the paste capillary porosity φ_p and the exponent ζ = 1.73 fitted to the Life-365 relation $D_{28} = 10^{(-12.06 + 2.40\ w/cm)}$ m²/s for OPC over w/c = 0.30–0.60 (model-to-Life-365 ratio within 1.23); D_wc applies the same Life-365 relation to the cement-only water-to-cement ratio, which treats filler as an inert diluent (pessimistic). Applying a mapping calibrated on OPC to filler-bearing pastes is an extrapolation, which is why both are retained. Corrosion initiates when C(x_cover, t) reaches a critical chloride content and a propagation period follows (Tuutti 1982).

Acid and leaching fronts. For a sharp reaction front, $x\, dx/dt = D_e\, C_a/N$ gives $x = \sqrt{(2 \int D_e\, dt\, C_a/N)}$, where C_a is the aggressive-species concentration and N the capacity of the matrix per unit volume. For acid attack, $N = 2[\xi_N \cdot 11.6\, \alpha\, c + \chi(d)\, m_{calcite}/0.10009]$ (mol H^+/m^3), where cement supplies 11.6 mol Ca per kg, ξ_N is the accessible share, and calcite supplies 2 mol H^+ per mole but only when particles are

fine enough: $\chi(d) = 1/[1 + (d/d_c)^2]$. For soft-water leaching, $N = \xi_L \cdot 11.6\ \alpha\ c$ (calcite does not contribute because of its low solubility), and a pressure-driven term $x_a = K\ i\ C_{sat}\ t/N$ is added in quadrature to the diffusive front, with hydraulic conductivity $K = \kappa_K\ D_e$. Mass loss from dissolved calcite is not tracked.

3.3 Damage accumulation

Total damage is the sum of a mechanical state D_m integrated in time and chemical and abrasion terms that are explicit functions of time:

$$dD_m/dt = n_{traf} / N_f(S_{eff}) + n_{ft}\ d_{ft}(S_w) \quad (6)$$

$$D(t) = D_0 + D_m(t) + D_f\ x_{chem}(t)/x_{lim} + D_f\ w_{abr}(t)/w_{lim} \quad (7)$$

Fatigue life follows the Aas-Jakobsen relation $\log_{10}N_f = (1 - S)/[0.0685(1 - R)]$ with stress ratio $R = 0.1$, and the effective stress ratio $S_{eff} = (\sigma_{traf} + \sigma_{env})/[f_r(t)(1 - D)]$ includes the continuum-damage amplification $1/(1 - D)$, so that damage accelerates as the section degrades. The traffic stress σ_{traf} is a fixed fraction (low, medium, high stress ratio class) of the OPC reference flexural strength, so that shell and OPC mixes carry the same absolute load; σ_{env} is the restrained shrinkage/curling stress. Freeze–thaw damage per cycle is $d_{ft} = d_1[(S_w - S_{crit})/(1 - S_{crit})]^q$ for $S_w > S_{crit}$, with $S_w = (\varphi_{cap} + a_f\ a_{air})/(\varphi_{cap} + a_{air})$ the degree of saturation (Fagerlund 1977). The chemical term has the form $D_f\ x/x_{lim}$, i.e. the failure damage D_f is reached exactly when the front reaches its limit, which fixes the coefficient B of a generic chemical driver by dimensional consistency rather than leaving it free. Early-age damage D_0 is switched on by a logistic function of the cracking indices $I_{sh} = \max(\sigma_{sh}/f_t)$ for restrained shrinkage (pavement) or I_{th} for restrained thermal contraction (dam); a mass-concrete temperature drop is $\Delta T = \psi_T\ c\ Q_{cem}\ \alpha(5\ d)/(\rho\ c_p)$.

Service life is the earliest of: chloride initiation plus propagation, and D reaching D_f; it is censored at 100 years. Because OPC reference concrete often survives the horizon, non-inferiority is defined with a tie-break: when the OPC reference survives 100 years, the shell mix must also survive and its damage index at 100 years must not exceed the reference by more than 5% of D_f.

3.4 Carbon accounting

$$E_{total} = E_{binder} + E_{shell} + E_{aggregate} + E_{water/admix} + E_{construction} - E_{credits} \quad (8)$$

The functional unit is 1 m^3 of concrete in service; cradle-to-gate embodied carbon E_{gate} excludes construction, and the annualised value divides E_{total} by service life. Cement carries a footprint EF_{cem} (0.75–0.95 kg CO_2e/kg). For P1, shell powder carries $EF_{sh} = [e_{dry}\ EF_{th} + E_g(d)\ EF_{el}]/1000 + (\text{distance} \times EF_{truck})/1000 - c_{land}$ per kg, where grinding energy scales as $E_g = E_{ref}\ (45\ \mu m/d)^{0.5}$ and c_{land} is a small landfill-avoidance credit. For P2 the change in cement footprint per kg is $\Delta EF = \rho\ m_{CaCO_3}\ [EF_{proc} - EF_{lq} + EF_{kiln}] - \beta_{bio}\ \rho\ m_{CaCO_3} \times 0.4397$, where 0.4397 kg CO_2/kg $CaCO_3$ is the stoichiometric calcination emission and β_{bio} is the fraction of that emission credited as biogenic. **The base case sets $\beta_{bio} = 0$**: shell calcium carbonate does not eliminate the CO_2 released on calcination. Organic CO_2 is treated as biogenic.

3.5 Falsification region

For uncertain parameters θ, let E be embodied carbon and L service life. The shell alternative fails to deliver a simultaneous advantage over its OPC reference, on the same draw of θ, in the set

$$\mathcal{F} = \{\ (r, \theta) : E_{shell} \geq E_{OPC}\ \vee\ L_{shell} \text{ not non-inferior to } L_{OPC}\ \} \quad (9)$$

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The probability P($\mathscr{F}$) is estimated by Monte Carlo and its complement P_adv is reported. Because carbon differences of a fraction of a percent are within the noise of the inputs, a second, material version requires E_shell ≤ 0.95 E_OPC and is reported as P_adv,mat. Both are given; the falsification maps use the material version.

4. Numerical methodology

4.1 Cases, parameters and sampling

Three cases are modelled with case-specific mix designs, exposures and mechanisms (Appendix A): (A) highway rigid pavement (fatigue, shrinkage, freeze–thaw, chloride from de-icing salt), (B) dam mass concrete (early-age thermal cracking, soft-water leaching with hydrostatic gradient, freeze–thaw), and (C) marine culvert / riverbed (chloride, acid, abrasion). The parameter study varies shell replacement (r = 10, 25, 50, 75, 100%), median particle size (10, 30, 100, 200 µm), chemical exposure (mild, moderate, severe), mechanical stress ratio (low, medium, high), relative humidity (55–100%), temperature (5–60 °C curing in the hydration study) and horizon (1–100 yr). The P2 pathway is varied over ρ = 25–100% and β_bio = 0, 50, 100%. For P1, two design rules are compared: **same w/b** (shell replaces cement at constant water and binder content, as in the source white paper) and **strength-matched** (binder content is increased at constant water until the 28-day strength equals the OPC reference; the search is restricted to the range in which the model strength increases with binder, and draws where matching is infeasible are flagged; 96% of draws are matchable at r = 25%).

Uncertain inputs (65 common and up to 11 case-specific parameters) are drawn by Latin hypercube sampling (McKay et al. 1979) with n = 3000. Every shell configuration is evaluated on the same draws as its OPC reference (common random numbers), so differences are attributable to the material and not to sampling noise. Global sensitivity uses Sobol total-order indices with the Jansen estimator (Saltelli et al. 2010; Sobol 2001), n = 2048 base samples with bootstrap intervals. The cases are time-stepped at 0.125 yr; hydration, chloride and front solutions are closed form or cumulative integrals.

4.2 Baseline anchoring and calibration

Three calibration steps use only the OPC reference. (i) The porosity–diffusivity exponent is fitted to the Life-365 relation (ζ = 1.73; maximum deviation factor 1.23). (ii) The strength model is set so that nominal OPC strength at w/c = 0.45 is about 43 MPa and the strength–w/c slope resembles an Abrams-type curve. (iii) Traffic cycles, curling stress, stress-ratio classes, surface chloride and cover are chosen so that the OPC reference reaches plausible design lives (median 24.8 yr for the pavement, 18.3 yr for the culvert, 98% survival to 100 yr for the dam). The last step anchors the absolute scale of the service life, so results should be read as **relative to the reference**, not as absolute life predictions. Shell-specific parameters are not calibrated in these three steps; they are tested and calibrated against published data in Section 4.4.

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4.3 Verification and validation

Table 3. Numerical verification (python ShellConcreteMC_v2.py --verify).

Test	Result
Baseline recovery: r = 0 (P1) and ρ = 0 (P2) reproduce OPC life and carbon	max difference 0
Closed-form hydration vs fourth-order Runge–Kutta integration	max error 3.6e-6
Time-varying-D chloride solution vs Crank–Nicolson finite differences	max error 2.2e-5 of C_s
Damage time-step convergence (Δt = 0.125 vs 0.0625 yr)	median 0.2%; 95th percentile 2.5% difference in life
Carbon budget closure (sum of parts = total)	residual 0
Calcination stoichiometry	0.4397 kg CO_2/kg $CaCO_3$
Strength matching reproduces the OPC 28-day strength (matched draws)	max relative error 1.3e-11; 97% feasible at r = 25%
D_{28} vs Life-365 relation, OPC, w/c = 0.30–0.60	within a factor 1.23

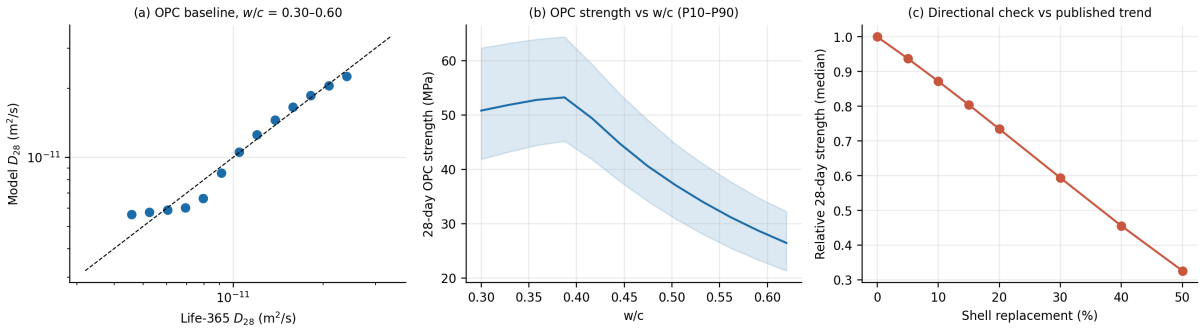


Fig. 2. Baseline checks. (a) Model D_{28} against the Life-365 relation for OPC. (b) Model OPC 28-day strength against w/c (P10–P90). (c) Median relative 28-day strength against shell replacement (P1, 25 µm).

Verification is not validation. The checks above show that the equations are solved correctly and that the baseline is recovered; they say nothing about whether the model describes shell concrete. The model is compared with published directional behaviour (Table 4) and, in Section 4.4, with the only open quantitative dataset found. Conformity with directional trends is a necessary, not sufficient, condition.

Table 4. Directional checks of the model against published trends for ground shell as cement replacement.

Published trend	Model check	Result
Strength decreases monotonically with replacement (5–50%)	P(strength decreases at every step), r = 0–50%	100%
Strength relative to OPC at 10 / 20 / 50% replacement	Median relative 28-day strength	0.87 / 0.73 / 0.33
Porosity / water absorption increases with replacement	P(capillary porosity at 50% > at 0%)	100%
Drying shrinkage decreases (some shell types)	P(shrinkage proxy at 20% < at 0%)	100%
Setting / early hydration retarded by shell organics	P(α(1 d) lower; unwashed shell, no filler effect)	99%

4.4 Out-of-sample test and calibration against published concrete data

Validation needs measured data. Of the open-access sources available to the author in full text, one reports tabulated constant-binder mixes with ground shell powder: Abbas and Kudukkan (2025), who used scallop

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shell as filler and partial cement replacement at a binder content of 350 kg/m³ and a water content of 160 kg/m³. The three mixes used here contain 50, 95 and 140 kg/m³ of shell-derived powder (r = 14%, 27% and 40% of the binder) with the median particle sizes reported there. The observed 28-day strengths relative to the control are read from the text for two mixes ("almost the same", "nearly 25% reduction") and derived for the third from the reported moduli of elasticity ($E \propto \sqrt{f_c}$); they are therefore approximate, and a tolerance of ±0.07 on the relative strength was used to represent this and specimen scatter.

Table 5. Out-of-sample test of the strength model against Abbas and Kudukkan (2025): 28-day compressive strength relative to the control mix, median (P10–P90) of model draws. Observed values: S0F15 and S30F15 from the text of the source, S15F15 derived from the reported moduli. The strength model is shared by all three cases.

Mix	Shell powder, r	Median size (µm)	Observed	Original model ($k_f = 0$)	With k_f prior	Calibrated (posterior)	Role
S0F15	14%	90	1.00	0.79 (0.75–0.83)	0.92 (0.81–1.03)	0.93 (0.93–0.94)	calibration
S15F15	27%	64	0.86	0.61 (0.54–0.68)	0.85 (0.65–1.05)	0.87 (0.87–0.88)	held out
S30F15	40%	55	0.75	0.43 (0.36–0.52)	0.77 (0.49–1.07)	0.81 (0.79–0.82)	calibration

The original model, in which shell powder is an inert diluent, predicts relative strengths of 0.79, 0.61 and 0.43 against 1.00, 0.86 and 0.75 observed; none of 100,000 prior draws falls within the tolerance for any mix. The inert-filler strength model is therefore rejected by these data: it over-predicts the strength lost to replacement, most clearly at the highest replacement (0.43 against 0.75).

A filler efficiency k_f (Section 3.1) was added with all other priors unchanged. Rejection approximate Bayesian computation was run on S0F15 and S30F15 (300,000 draws, tolerance 0.07 on each relative strength) and accepted 10,999 draws (3.7%); S15F15 was held out. The posterior median of k_f is 0.65 (P10–P90 0.54–0.84): a unit mass of shell powder contributes roughly two thirds of the strength-giving capacity of cement at these dosages. The posterior predictive for the held-out mix is 0.87–0.88 against 0.86 observed, and 100% of accepted draws lie within the tolerance (Fig. 3a). Two limits of the test must be stated. First, S15F15 lies between the two calibration mixes in replacement level, so this is an interpolation and not an extrapolation to higher replacement or to oyster shell. Second, the calibrated model sits at the edge of the tolerance for S0F15 (0.93 against 1.00) and above the observation for S30F15 (0.81 against 0.75), so a bias of 0.06–0.07 remains in the strength prediction; the fitted strength–replacement curve is flatter than the two calibration points require, which indicates a residual structural misfit of the strength relation. Apart from k_f the shell parameters are only weakly constrained (Fig. 3b).

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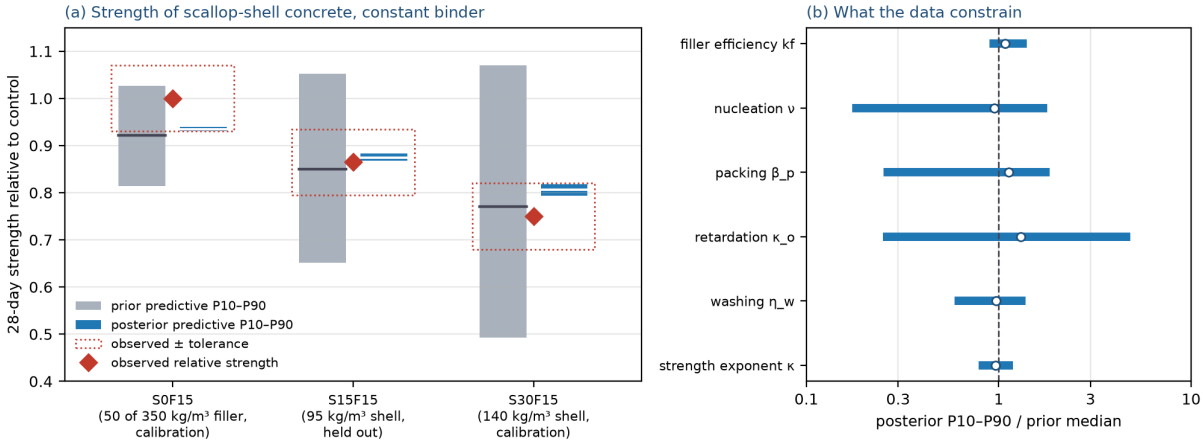


Fig. 3. Out-of-sample test against Abbas and Kudukkan (2025). (a) Relative 28-day strength: prior predictive of the original model structure with the k_f prior (grey), calibrated posterior predictive (blue) and observed values with the ±0.07 tolerance (red); S15F15 was not used in the calibration. (b) Posterior 10th–90th percentile range of selected parameters relative to their prior medians (circle: posterior median).

The carbon accounting was checked against the same paper. With its emission factors (cement 0.91 kg CO_2e/kg, shell powder 0.059 kg CO_2e/kg) the model arithmetic gives −24.4% and −36.0% for S15F15 and S30F15 relative to the control; this is the same budget the model uses and shows only that it is consistent, not an independent test. The model's own shell emission factor at 55 µm, 0.058 kg CO_2e/kg (P10–P90 0.036–0.084), is of the same order as the laboratory value of 0.059, but the system boundary of that value is not stated here and the agreement should not be over-read.

The calibration constrains strength only. Porosity, and hence chloride diffusivity, count cement only in the model and k_f is not applied to transport, so the calibrated posterior changes strength-related quantities, fatigue and strength-matched designs but not directly the chloride, acid or freeze–thaw mechanisms. No open dataset with tabulated chloride, acid or freeze–thaw results for shell-powder concrete was found, and the transport and damage predictions therefore remain uncalibrated.

5. Results

Sections 5–7.3 report the original model with inert shell filler (k_f = 0), which Section 4.4 shows to over-predict the strength lost to replacement. Section 7.4 re-evaluates the main claims under the calibrated posterior, and the two sets of results should be read together.

5.1 Hydration and microstructure

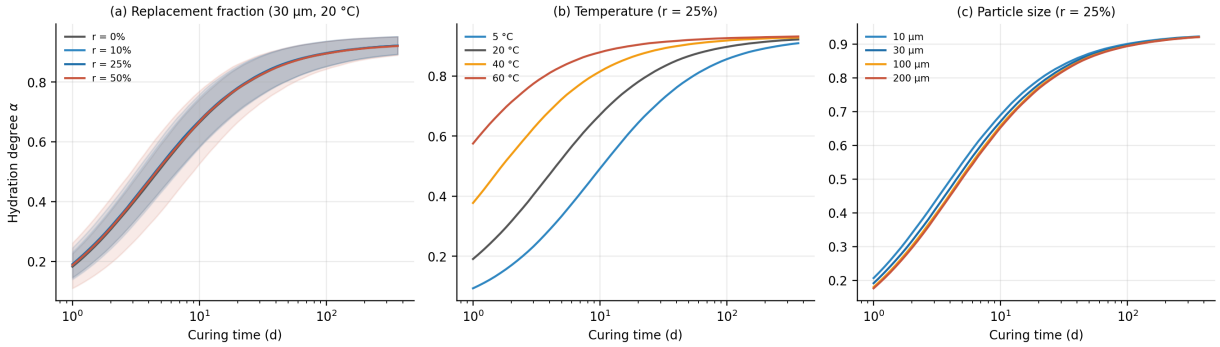


Fig. 4. Hydration degree against curing time (median and P10–P90): (a) shell replacement fraction; (b) curing temperature; (c) median particle size (P1, w/b = 0.42).

The model does not predict a large acceleration. Median hydration degree at 1 day is 0.18 for OPC and 0.19, 0.19 and 0.18 for 10%, 25% and 50% replacement; at 28 days the values are 0.81, 0.82, 0.82 and 0.81. The nucleation effect and the larger effective w/c are largely offset by organic retardation and dilution, so the probability that 1-day hydration is higher than OPC at r = 20% is 69% (the sign depends on the poorly known filler and retardation coefficients). Meanwhile cement is removed: median 28-day strength falls to 66% of OPC at r = 25% and 32% at r = 50%, and capillary porosity per unit paste rises by 46% and 94%. Temperature (Fig. 4b) accelerates hydration through the Arrhenius factor, and finer shell (Fig. 4c) accelerates it moderately.

5.2 Transport

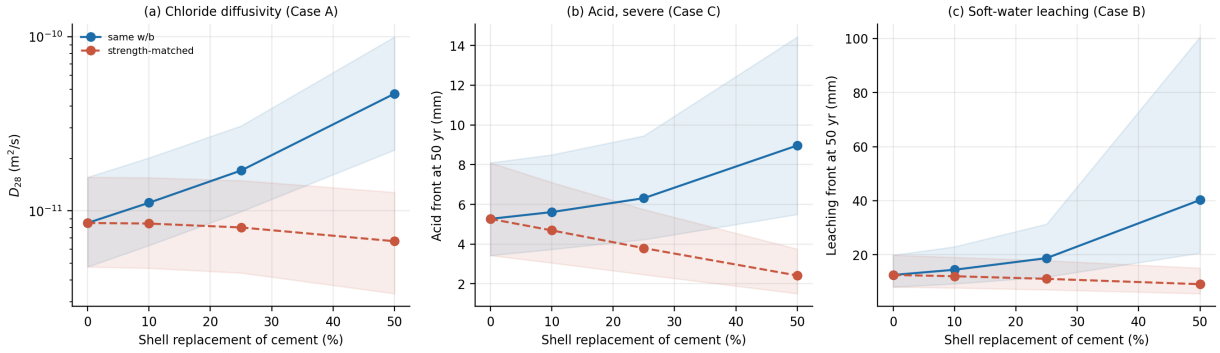


Fig. 5. Transport indicators against shell replacement for same-w/b and strength-matched mixes (median and P10–P90): (a) 28-day chloride diffusivity, pavement; (b) acid front depth at 50 years, severe exposure, culvert; (c) leaching front depth at 50 years, dam.

At constant w/b the median 28-day chloride diffusivity rises by a factor 1.3 at r = 10%, 2.1 at 25% and 5.7 at 50%, because dilution raises porosity and the cement-only mapping amplifies the effect. In the severe acid case the 50-year front is 5.3 mm for OPC and 6.3 mm at r = 25% (same w/b, +20%) but 3.8 mm when strength-matched (−28%): calcite buffering helps only when it is not outweighed by dilution. Particle size matters: at r = 25% the front is 6.2 mm for 10 µm shell and 7.3 mm for 200 µm shell, because coarse particles are less accessible. For soft-water leaching the 50-year front is 12.6 mm for OPC, 19.0 mm at r = 25% (+51%) and 40.6 mm at r = 50%, because calcite adds no leachable calcium and dilutes the cement. The calcite-dissolution mass loss in acid, which could offset the buffering benefit, is not modelled.

5.3 Service-life distributions

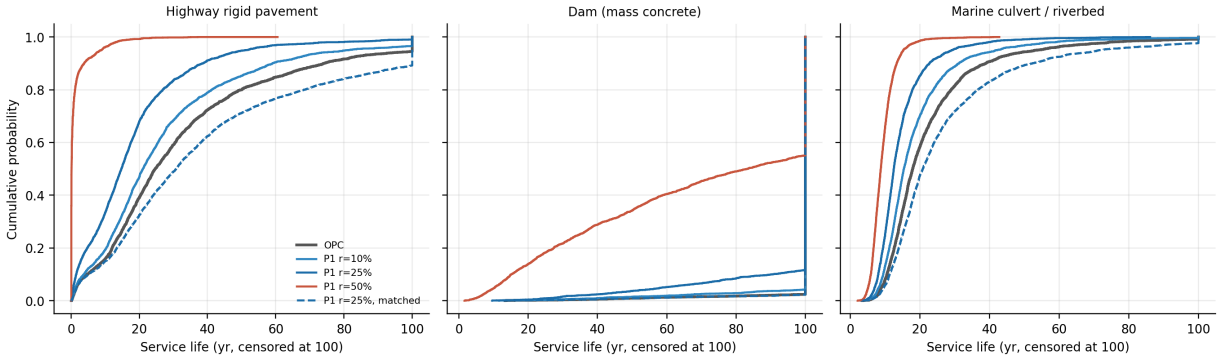


Fig. 6. Empirical cumulative distribution of service life (censored at 100 years) for OPC and selected shell mixes in the three cases (medium load, moderate exposure; 30 µm shell).

The OPC reference has a median life of 24.8 yr (P10–P90 4.2–74.3) in the pavement case, 18.3 yr (10.7–38.4) in the culvert case, and survives 100 years in 98% of dam draws. At constant w/b, shell replacement shifts the distributions to shorter life in the pavement and culvert cases, and strongly so at r ≥ 50% (Fig. 6). Strength-matched mixes shift to longer life, but this benefit rests on the lower w/b implied by the added binder and on the diffusivity mapping (Section 7.3). Details by case are in Section 6.

5.4 Carbon

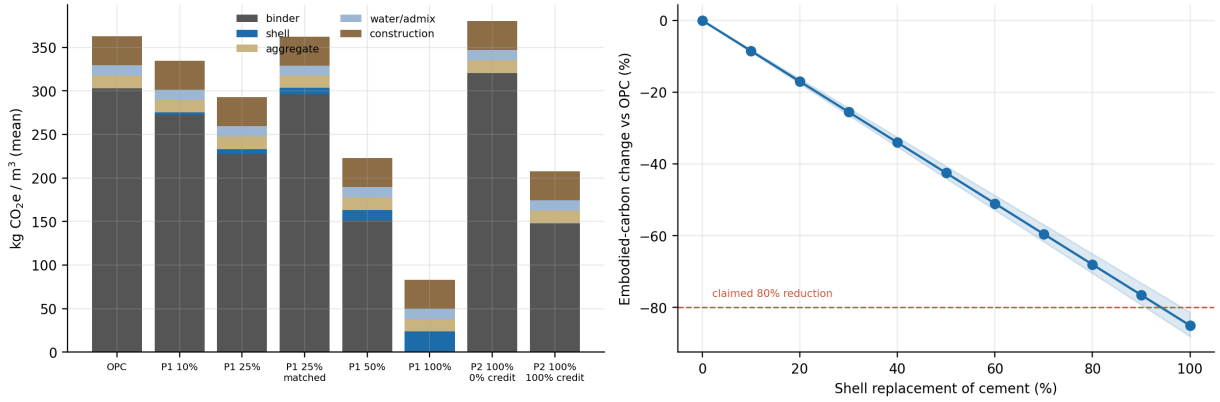


Fig. 7. Left: mean embodied carbon by component (pavement mix). Right: change in embodied carbon against shell replacement at constant w/b (median and P10–P90); the dashed line marks the 80% reduction claimed in the source white paper.

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Table 6. Change in cradle-to-gate embodied carbon relative to OPC (median, P10–P90) and relative 28-day strength; pavement mix, 30 µm shell.

Configuration	Carbon change vs OPC (%)	28-day strength vs OPC
P1, r = 10%, same w/b	−8.5 (−8.8 to −8.2)	0.87
P1, r = 25%, same w/b	−21.2 (−22.0 to −20.4)	0.67
P1, r = 50%, same w/b	−42.5 (−44.0 to −40.8)	0.33
P1, r = 100%, same w/b	−85.0 (−88.0 to −81.5)	0.00
P1, r = 25%, strength-matched	−0.3 (−3.0 to +3.1)	1.00
P1, r = 50%, strength-matched	+3.6 (−1.5 to +10.9)	0.99
P2, ρ = 100%, no biogenic credit	+5.1 (+2.8 to +8.1)	n/a (≈ 1.0)
P2, ρ = 100%, 50% biogenic credit	−21.0 (−24.1 to −17.8)	n/a (≈ 1.0)
P2, ρ = 100%, 100% biogenic credit	−47.1 (−51.8 to −42.7)	n/a (≈ 1.0)

At constant w/b, carbon falls almost proportionally to r (−8.5%, −21.2%, −42.5% at 10, 25 and 50%) because the shell powder footprint is small relative to cement. The claimed 80% reduction is reached only at essentially complete cement replacement (−85.0% at r = 100%), at which the concrete has no strength. When strength is matched by adding binder, the benefit vanishes (−0.3% at r = 25% and +3.6% at r = 50%). Pathway P2 increases embodied carbon by 5.1% (P10–P90 2.8 to 8.1) when no biogenic credit is allowed, because processing and transport of shell exceed the footprint of quarrying adjacent limestone while the calcination CO_2 is unchanged; a reduction appears only when 50% or 100% of the calcination CO_2 is credited (-21.0% and -47.1%), which is an accounting choice and not a material property.

5.5 Carbon–durability trade-off

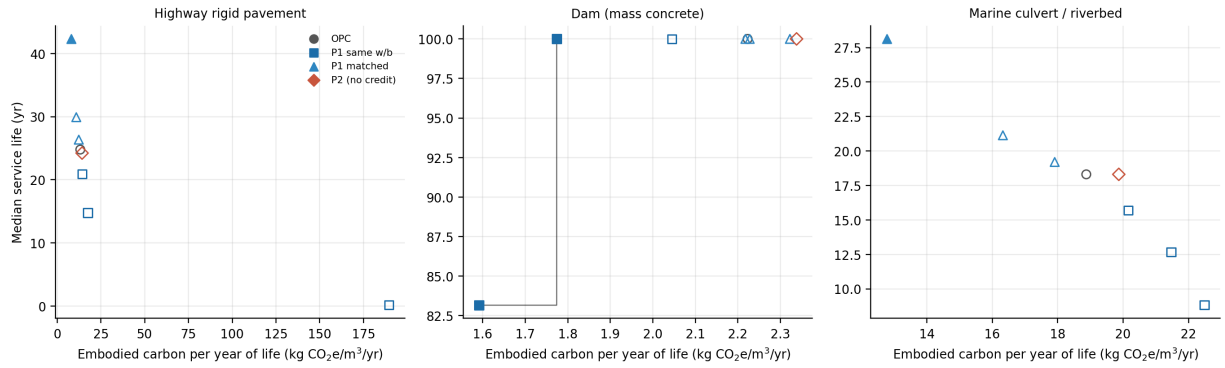


Fig. 8. Embodied carbon per year of life against median service life for OPC, P1 (r ≤ 50%, 30 µm; same w/b and strength-matched) and P2 without credit. Filled symbols are Pareto-optimal, open symbols are dominated.

Over the plausible replacement range (r ≤ 50%), the Pareto set is formed by strength-matched P1 mixes in the pavement and culvert cases (longer life at near-unchanged carbon per m³) and, in the dam case, by low-replacement mixes whose life is bounded by the 100-year horizon (Fig. 8). Same-w/b mixes are dominated in the pavement and culvert cases. These are median-based statements; the joint-advantage probabilities in Section 7 give the more informative picture.

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6. Infrastructure applications

6.1 Case A: highway rigid pavement

The pavement case combines repeated traffic loading, restrained shrinkage and curling, freeze–thaw and chloride ingress from de-icing salt (medium stress ratio, moderate exposure). The OPC failure mode is chloride corrosion initiation plus propagation in 72% of draws and damage in 23% (Table 7).

Table 7. Case A results (30 µm shell; 3,000 paired draws; medium stress ratio, moderate chloride exposure).

Mix	f_c28 (MPa)	Life P10 / P50 / P90 (yr)	ΔL P50 (yr)	P(non-inferior)	ΔE P50 (%)	P(≥5% cut & non-inf.)
OPC reference	49	4.2 / 24.8 / 74.3	–	–	–	–
P1 10%, same w/b	42	3.3 / 20.9 / 59.4	−3.1	4%	−8.5	4%
P1 25%, same w/b	32	1.4 / 14.8 / 38.2	−9.0	1%	−21.2	1%
P1 50%, same w/b	16	0.1 / 0.1 / 4.5	−23.3	0%	−42.5	0%
P1 10%, matched	48	4.2 / 26.4 / 81.7	+1.1	88%	−0.5	0%
P1 25%, matched	48	4.4 / 29.9 / 100.0	+3.5	90%	−0.3	1%
P1 50%, matched	48	4.8 / 42.3 / 100.0	+11.2	92%	+3.6	1%
P2 100%, no credit	47	3.7 / 24.2 / 72.8	0.0	5%	+5.1	0%
P2 100%, 50% credit	47	3.7 / 24.2 / 72.8	0.0	5%	−21.0	5%

At constant w/b, replacement reduces median life by 3.1 yr (10%) and 9.0 yr (25%), and at 50% most draws fail within a year, since the weaker, more porous paste fatigues and admits chloride faster. The restrained-shrinkage cracking index exceeds 1 in 45% of OPC draws and in 37%, 25% and 5% of draws at 10%, 25% and 50% replacement (same w/b), a genuine shrinkage benefit that arises from dilution; it disappears when strength is matched (44%). Strength-matched mixes lengthen median life by 3.5 yr at r = 25%, but the carbon is unchanged. Sensitivity of the life difference is led by restraint factor (0.28), chloride-diffusivity multiplier (0.27), curling-stress factor (0.25).

6.2 Case B: dam (mass concrete)

For mass concrete the modelled mechanisms (thermal cracking, leaching with hydrostatic gradient, freeze–thaw on the face) rarely end the life of the OPC reference within 100 years (98% survive), so differences in life are censored and the damage-index tie-break governs (Table 8). Lower cement content reduces the adiabatic temperature rise, but the OPC thermal cracking probability is already 0%, so little is gained. At r = 10% (same w/b) the probability of non-inferior life is 57% and that of a joint 5% carbon cut and non-inferior life is 57%; at 25% it falls to 4% because the leaching front deepens (Section 5.2), and at 50% it is 0%. The apparent success at low replacement is therefore a statement that dam concrete is durable in the model regardless, not evidence of benefit, and the reference here is OPC although dam concrete usually contains supplementary cementitious materials. Life differences are driven by air-void filling fraction (0.38), critical degree of saturation (0.31), permeability factor (0.29).

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Table 8. Case B results (30 µm shell; 3,000 paired draws; moderate exposure). Median life is censored at 100 years for the OPC reference.

Mix	f_c28 (MPa)	Life P10 / P50 / P90 (yr)	ΔL P50 (yr)	P(non-inferior)	ΔE P50 (%)	P(≥5% cut & non-inf.)
OPC reference	34	100.0 / 100.0 / 100.0	–	–	–	–
P1 10%, same w/b	29	100.0 / 100.0 / 100.0	0.0	57%	−8.1	57%
P1 25%, same w/b	22	93.9 / 100.0 / 100.0	0.0	4%	−20.2	4%
P1 50%, same w/b	10	16.4 / 83.2 / 100.0	−15.8	0%	−40.4	0%
P1 10%, matched	34	100.0 / 100.0 / 100.0	0.0	99%	−0.5	0%
P1 25%, matched	34	100.0 / 100.0 / 100.0	0.0	98%	−0.2	0%
P1 50%, matched	34	100.0 / 100.0 / 100.0	0.0	99%	+3.5	0%
P2 100%, no credit	33	100.0 / 100.0 / 100.0	0.0	100%	+4.9	0%
P2 100%, 50% credit	33	100.0 / 100.0 / 100.0	0.0	100%	−20.0	100%

6.3 Case C: marine culvert / riverbed

The culvert case combines chloride ingress, acid from organic matter and abrasion. OPC failure is by chloride corrosion in 96% of draws (Table 9). At constant w/b the median life falls by 2.5 yr at 10%, 5.7 yr at 25% and 9.4 yr at 50%, and the probability of non-inferior life is 0%, 0% and 0%: shell concrete underperforms OPC in every draw when strength is not matched. Strength matching changes the sign of the median effect (+2.6 yr at 25%), with the joint-advantage probability at most 1%. The acid benefit of calcite buffering (Section 5.2) is too small relative to chloride-driven life to alter the failure mode. Life differences are driven by chloride-diffusivity multiplier (0.38), surface chloride (0.16), air-void filling fraction (0.15).

Table 9. Case C results (30 µm shell; 3,000 paired draws; moderate exposure).

Mix	f_c28 (MPa)	Life P10 / P50 / P90 (yr)	ΔL P50 (yr)	P(non-inferior)	ΔE P50 (%)	P(≥5% cut & non-inf.)
OPC reference	49	10.7 / 18.3 / 38.4	–	–	–	–
P1 10%, same w/b	42	9.7 / 15.7 / 31.1	−2.5	0%	−8.5	0%
P1 25%, same w/b	32	8.3 / 12.7 / 22.2	−5.7	0%	−21.4	0%
P1 50%, same w/b	16	5.9 / 8.8 / 13.2	−9.4	0%	−42.7	0%
P1 10%, matched	48	11.1 / 19.2 / 41.8	+0.7	89%	−0.5	0%
P1 25%, matched	48	11.8 / 21.1 / 51.6	+2.6	93%	−0.3	1%
P1 50%, matched	48	13.5 / 28.1 / 87.0	+9.2	96%	+3.7	1%
P2 100%, no credit	47	10.7 / 18.3 / 38.4	0.0	97%	+5.1	0%
P2 100%, 50% credit	47	10.7 / 18.3 / 38.4	0.0	97%	−21.1	97%

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Table 10. Sensitivity to exposure severity and load class for P1 at r = 25% (30 μm): median life and P(≥5% carbon cut and non-inferior life), same w/b and strength-matched.

Case and condition	OPC life P50	Same w/b: life P50	P(adv, mat)	Matched: life P50	P(adv, mat)
A: moderate exposure, low load	29.4	19.7	0.05	36.9	0.01
A: moderate exposure, medium load	25.1	14.5	0.01	30.4	0.01
A: moderate exposure, high load	14.3	1.5	0.01	15.7	0.01
A: mild exposure, medium load	50.2	22.5	0.08	63.8	0.01
A: severe exposure, medium load	17.3	11.4	0.01	20.2	0.01
B: mild exposure, medium load	100.0	100.0	0.14	100.0	0.00
B: moderate exposure, medium load	100.0	100.0	0.05	100.0	0.00
B: severe exposure, medium load	100.0	100.0	0.02	100.0	0.00
C: mild exposure, medium load	31.6	19.4	0.00	38.7	0.01
C: moderate exposure, medium load	18.4	12.7	0.00	21.4	0.01
C: severe exposure, medium load	13.8	10.4	0.00	15.5	0.01

Across exposure severity and load class the qualitative conclusion is unchanged: no condition yields a high probability of joint advantage for P1 at r = 25%. Higher stress ratio and harsher exposure shorten the OPC life and compress the differences, while milder conditions leave the shell mix further behind in relative terms.

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7. Uncertainty and falsification

7.1 Falsification maps

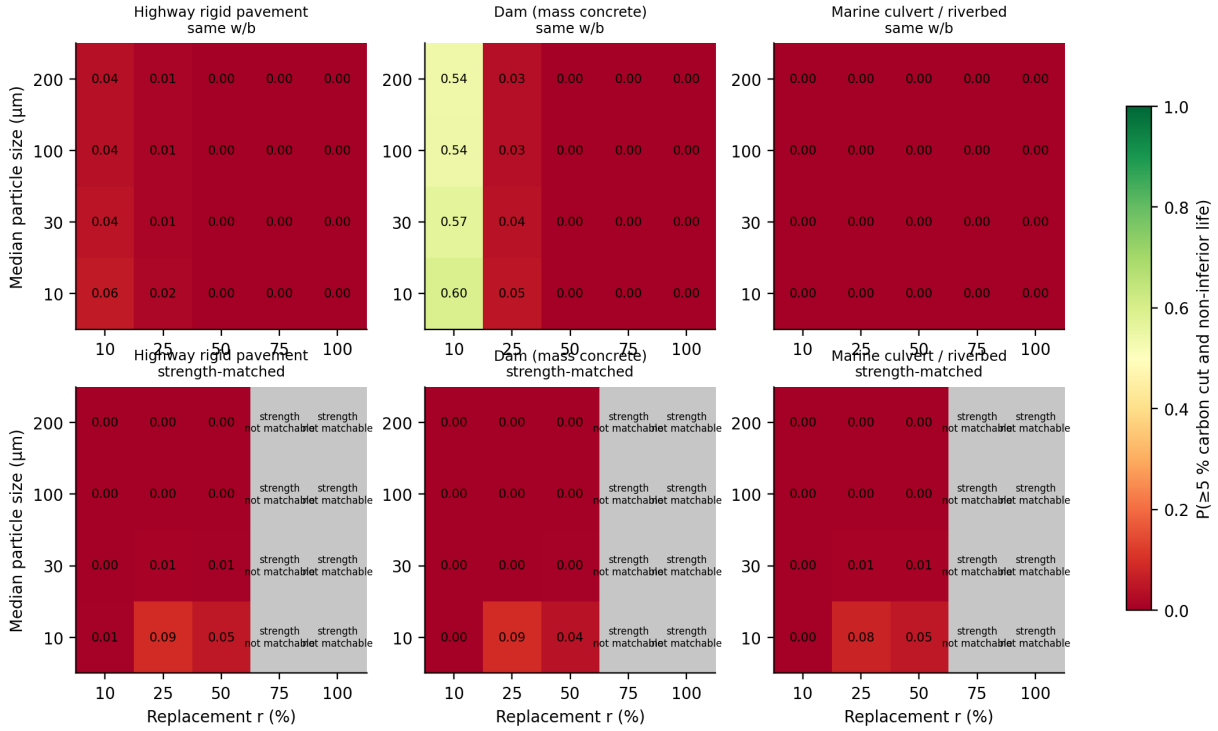


Fig. 9. Probability of a joint advantage (carbon cut of at least 5% and non-inferior service life) for P1 as a function of replacement fraction and median particle size, same-w/b (top) and strength-matched (bottom) designs. Grey cells mark strength-matched designs for which matching is infeasible in more than half of the draws.

The joint-advantage region is small. For the pavement and culvert cases the same-w/b maps are essentially empty (maximum for r ≤ 50%: 0.06 and 0.00), and the strength-matched maps are empty wherever matching is possible, because the carbon benefit is below 5%. Fine shell (10 µm) at r = 25–50% gives the largest matched-design values (up to 0.09 for the pavement), because the stronger filler effect of fine shell (higher nucleation and packing coefficients) means less binder has to be added to restore strength, and calcite is more accessible to acid; finer grinding itself costs more energy. The dam map is non-empty at r = 10% only for the censoring reason explained in Section 6.2. With the strict definition (any carbon reduction), matched mixes appear to succeed in about half of the draws ($P_{adv} = 0.48$ for the pavement, r = 25%), but the carbon change is only −0.3% (P10–P90 −3.0 to +3.1%), which is within the spread produced by the inputs, and is why the material threshold is used.

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7.2 Pathway P2

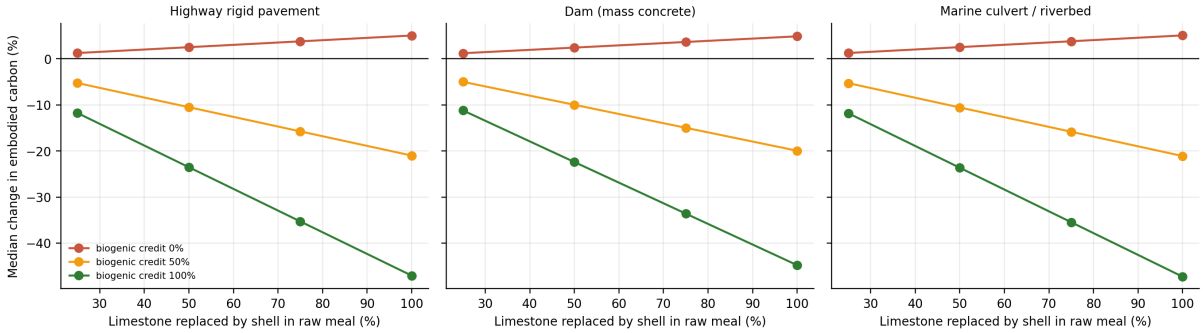


Fig. 10. P2 pathway: median change in embodied carbon against the fraction of limestone replaced by shell in raw meal, for three levels of biogenic credit.

For P2 the durability effect is negligible (median life change −0.04 yr in the pavement case from the impurity penalty, none in the dam and culvert cases), so the outcome is determined by carbon accounting. Without credit P2 is falsified: embodied carbon is higher in 100% of draws. With a 50% or 100% biogenic credit, carbon falls by 21.0% and 47.1% and the joint advantage is reached in 97%–100% of dam and culvert draws (for the pavement, the strict non-inferiority test fails in most draws because of the small strength penalty; with a 5% life tolerance the probability is 81%). Whether shell calcination CO_2 may be treated as biogenic is a methodological and regulatory question that this model cannot resolve; the credit should be reported as a scenario, not a property.

7.3 Model-form uncertainty and sensitivity

Table 11. Dependence of the median life difference (ΔL, years) for P1 at r = 25% on the diffusivity model weight χ_D (0 = porosity mapping, 1 = cement-only w/c mapping).

χ_D range	Pavement: same w/b	Pavement: matched	Culvert: same w/b	Culvert: matched
0.00–0.25	−8.0	+5.7	−5.3	+4.5
0.25–0.50	−9.1	+4.7	−5.5	+3.2
0.50–0.75	−9.0	+3.2	−5.5	+2.2
0.75–1.00	−10.1	+2.1	−5.8	+1.2

The sign of the same-w/b effect is robust to the diffusivity mapping (negative in all bins), whereas the size of the matched-design benefit declines from +5.7 yr to +2.1 yr in the pavement case as the pessimistic mapping is weighted more heavily. The matched-design benefit also depends on the assumed filler shift ω of the ageing exponent, which favours filler-bearing mixes and has no experimental support here.

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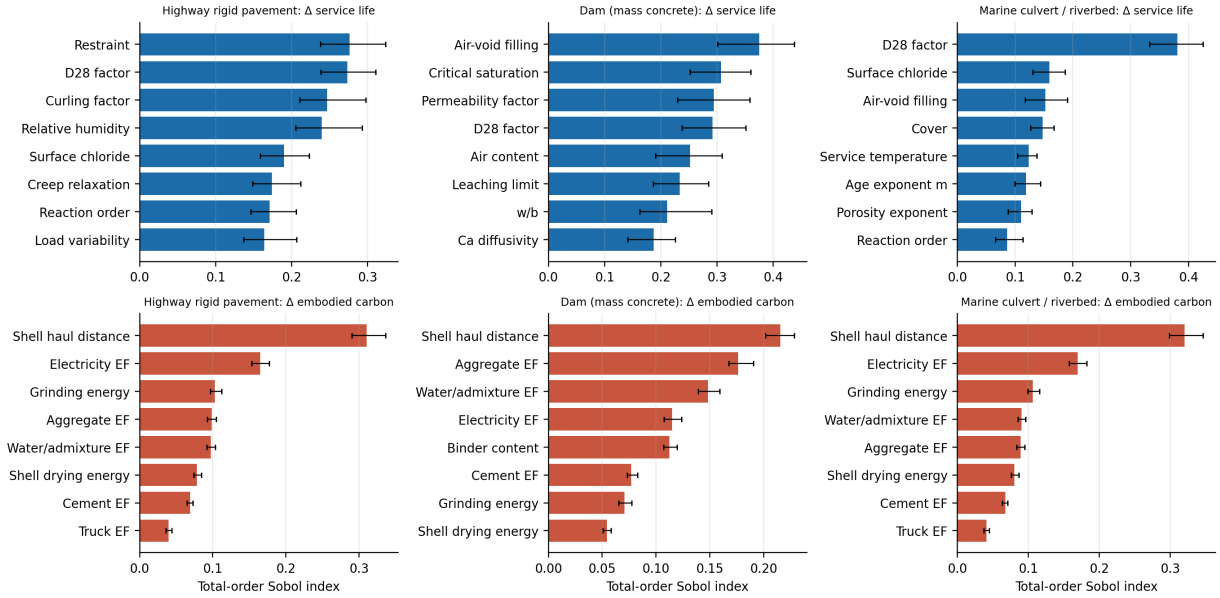


Fig. 11. Total-order Sobol indices (eight largest, with 95% bootstrap intervals) for the difference in service life (top) and in embodied carbon (bottom) between P1 at r = 25% and OPC, in each case.

The life difference is controlled by restraint factor (0.28), chloride-diffusivity multiplier (0.27), curling-stress factor (0.25), relative humidity (0.24) in the pavement case, by air-void filling fraction (0.38), critical degree of saturation (0.31), permeability factor (0.29), chloride-diffusivity multiplier (0.29) in the dam case, and by chloride-diffusivity multiplier (0.38), surface chloride (0.16), air-void filling fraction (0.15), cover depth (0.15) in the culvert case; the sums of the total-order indices exceed one, indicating interactions. The first-order Jansen estimates were noisy (the sums of first-order indices for the three life-difference analyses were 0.60, -0.49 and 0.01, whereas a stable estimate lies between 0 and 1) because of the many parameters and the censored, tie-heavy output, and are not reported. The carbon difference is controlled by logistics and energy inputs: shell haul distance (0.31), electricity emission factor (0.16), grinding energy (0.10). The shell haul distance alone explains 31% of the carbon variance for the pavement, which means that carbon claims for shell concrete cannot be separated from the supply-chain geography.

7.4 Claims re-evaluated under the calibrated filler efficiency

To show how far the conclusions depend on the strength model, the main configurations were re-evaluated on the 10,999 accepted draws (calibrated) and on prior draws with k_f = 0 (original), for 30 µm shell (Table 12, Fig. 12). The calibrated draws are conditioned on the strength data only.

Table 12. Original (k_f = 0) and calibrated results for P1, 30 µm shell: median carbon change, 28-day strength relative to OPC, median life difference, and P(carbon cut of at least 5% and non-inferior life).

Case and design	ΔE P50 (%), orig. / cal.	f_c28 vs OPC, orig. / cal.	ΔL P50 (yr), orig. / cal.	P(adv, mat), orig. / cal.
A: r = 10%, same w/b	−8.5 / −8.5	0.87 / 0.97	−3.1 / −2.2	0.05 / 0.27
A: r = 25%, same w/b	−21.2 / −21.2	0.66 / 0.91	−9.1 / −5.9	0.01 / 0.23
A: r = 25%, matched	−0.3 / −16.4	1.00 / 1.00	+3.8 / −3.3	0.01 / 0.28
A: r = 50%, same w/b	−42.5 / −42.5	0.33 / 0.77	−23.7 / −12.3	0.00 / 0.18
A: r = 50%, matched	+3.5 / −33.1	0.99 / 1.00	+12.1 / −7.9	0.01 / 0.24
B: r = 10%, same w/b	−8.1 / −8.1	0.86 / 0.96	0.0 / 0.0	0.56 / 0.55

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Case and design	ΔE P50 (%), orig. / cal.	f_c28 vs OPC, orig. / cal.	ΔL P50 (yr), orig. / cal.	P(adv, mat), orig. / cal.
B: r = 25%, same w/b	−20.2 / −20.2	0.64 / 0.90	0.0 / 0.0	0.05 / 0.04
B: r = 25%, matched	−0.1 / −15.4	1.00 / 1.00	0.0 / 0.0	0.00 / 0.19
B: r = 50%, same w/b	−40.4 / −40.5	0.30 / 0.75	−19.1 / −18.9	0.00 / 0.00
B: r = 50%, matched	+3.7 / −31.2	1.00 / 1.00	0.0 / 0.0	0.01 / 0.02
C: r = 10%, same w/b	−8.5 / −8.5	0.87 / 0.97	−2.5 / −2.5	0.00 / 0.00
C: r = 25%, same w/b	−21.3 / −21.4	0.66 / 0.91	−5.5 / −5.5	0.00 / 0.00
C: r = 25%, matched	−0.3 / −16.5	1.00 / 1.00	+2.6 / −4.0	0.01 / 0.00
C: r = 50%, same w/b	−42.7 / −42.7	0.33 / 0.77	−9.3 / −9.0	0.00 / 0.00
C: r = 50%, matched	+3.5 / −33.2	0.99 / 1.00	+9.3 / −7.2	0.01 / 0.00

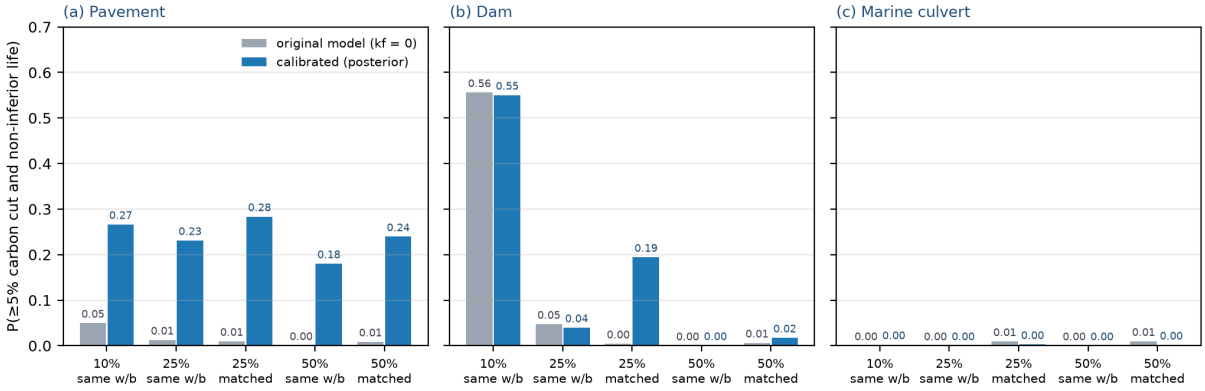


Fig. 12. Probability of a joint advantage (carbon cut of at least 5% and non-inferior life) for the original model (k_f = 0, grey) and under the strength-calibrated posterior (blue), for P1 with 30 µm shell.

Calibration changes the strength picture substantially: at r = 25% and constant w/b the strength is 0.91 of OPC rather than 0.66, and a strength-matched design needs little added binder, so the carbon cut of the matched mix is −16.4% instead of −0.3% (−33.1% at r = 50%). The joint-advantage probability for the pavement rises from 0.01 to 0.23 (same w/b) and from 0.01 to 0.28 (matched), and for the dam at r = 25% matched from 0.00 to 0.19. The original conclusion that a joint advantage is essentially unattainable is therefore not robust to the strength model.

Three features remain. First, even the calibrated matched designs have a median life difference of −3.3 yr in the pavement case (P10–P90 −15.8 to +7.4), because the transport model counts cement only in the paste porosity and so treats shell as inert for chloride; a joint advantage is reached in about one draw in four, not in most. Second, the culvert case stays at 0.00 or below in every configuration, so chloride-controlled marine exposure is not served by shell replacement in this model. Third, the dam result at r = 10% (0.55) again reflects the censoring of a reference that survives 100 years in the model rather than a demonstrated benefit. The two sets of results bracket the plausible range: the original model is rejected by the strength data, and the calibrated model is conditional on an untested transport mapping.

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8. Discussion

8.1 Physical interpretation

The main results follow from three mechanisms. First, shell powder is a diluent: removing cement lowers strength and raises porosity almost in proportion, so any benefit must come from a filler effect. The original coefficients make that effect small; the published constant-binder data imply a substantial filler efficiency (Section 4.4, $k_f \approx 0.65$), and this single parameter moves the joint-advantage probability in the pavement case from about 0.01 to about 0.23 (Section 7.4). Second, embodied carbon is dominated by cement, so replacing cement lowers carbon in proportion to r at constant w/b; at constant strength the added binder re-introduces the cement, and the carbon cut falls from about −21% to −16% at r = 25% under the calibrated filler efficiency and to almost nothing under the original one. Third, the carbon benefit of P2 is tied to a convention (biogenic credit), since the calcination CO_2 per kg of $CaCO_3$ is fixed by stoichiometry. The dam and culvert results show where shell concrete might be advantageous (low replacement in mass concrete, fine shell and matched strength in acid exposures) but these are small, conditional effects.

8.2 Assessment of the white paper's claims

Table 13. Source claims restated as hypotheses with model outputs and the validation each would need.

Claim	Model output (this study)	Status	Validation needed
Up to 80% CO_2 reduction	−85.0% only at r = 100% (no strength); −21.2% / −42.5% at r = 25 / 50% (same w/b); −0.3% / +3.6% matched; P2 +5.1% without credit	Not supported	Plant-specific cradle-to-gate LCA with defined boundary and functional unit
Thermal resistance to 2,400 °F	Not modelled	Not supported (see text)	Furnace tests with residual strength; thermogravimetry of the shell binder
100% limestone substitution	P2, ρ = 100%: carbon +5.1% (no credit); durability unchanged	Feasible chemically; no carbon benefit without credit	Raw-meal burnability, clinker chemistry, chloride and alkali limits
Accelerated hydration	α(1 d): 0.18 (OPC) vs 0.19 (r = 25%); strength 66% of OPC	Marginal acceleration; strength lower	Isothermal calorimetry; setting time; strength at 1–365 d
Reduced shrinkage	Cracking probability 45% → 25% at r = 25% (same w/b); unchanged when matched	Supported only as dilution	Restrained-ring and free-shrinkage tests
Superior acid resistance	50-yr front +20% (same w/b) vs −28% (matched); particle-size dependent	Conditional	Acid immersion with mass loss and depth of degradation
Higher structural durability	ΔL P50 −9.0 yr (pavement) and −5.7 yr (culvert) at r = 25% same w/b; +3.5 / +2.6 yr matched	Not supported at same w/b; conditional if matched	Chloride migration, freeze–thaw, fatigue and abrasion tests on matched mixes

On the thermal claim. 2,400 °F is about 1,316 °C. Calcite decomposes to CaO and CO_2 at roughly 900 °C (about 1,650 °F) at atmospheric CO_2 pressure (standard thermodynamic data; Chase 1998), so a calcium-carbonate-rich binder is not expected to withstand 2,400 °F; even conventional concrete suffers substantial strength loss and spalling at much lower temperatures. No model of fire behaviour is attempted here and the claim should be removed unless supported by formulation-specific tests.

8.3 Limitations

- No new experiments are reported. Shell-specific parameters (ν, β_p, κ_o, ω, k_{imp}) are literature-style ranges; only the strength model was calibrated, on one published scallop-shell dataset (three mixes, replacement 14–40% of the binder) whose strengths are partly read from text and derived from

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moduli, and the held-out mix lies between the calibration mixes, so extrapolation to higher replacement, to oyster shell and to other cements is untested. Rejection ABC accepted 3.7% of draws and constrains little besides k_f.

- Chloride, acid and freeze–thaw transport remain uncalibrated: paste porosity counts cement only, the filler efficiency is not applied to transport, and the claims that depend on life (Section 7.4) are therefore bracketed rather than determined.
- The OPC reference lives are anchored to plausible design lives by adjusting OPC-only parameters; results are relative, and the absolute lives are not predictions.
- The diffusivity mapping for filler-bearing pastes is an extrapolation of a relation calibrated for OPC; the two mappings bracket but do not validate it.
- The reference is plain OPC concrete; dam and marine concretes normally contain supplementary cementitious materials, which would change both carbon and durability baselines.
- Mechanisms are combined additively and one-dimensionally; carbonation, alkali–silica reaction, sulfate attack, reinforcement corrosion kinetics, calcite dissolution mass loss and the interaction of cracking with all transport processes are not modelled; surface chloride is constant.
- Carbon factors are generic screening values; the avoided-landfill credit, shell logistics and biogenic accounting are uncertain and dominate the carbon variance.
- The strength-matching rule (adding binder at constant water) is only one of several ways to achieve equal strength and is not applicable where the model strength saturates with binder.

8.4 Novelty and validation requirements

The contribution is not the finding that fine calcium carbonate acts as a filler, which is established for limestone, but a framework that couples processing-dependent carbon with hydration, transport and damage, evaluates each shell mix against its own OPC reference under shared uncertainty, and reports where the joint advantage fails. To be a validated model, the following would be needed: (i) isothermal calorimetry, porosimetry and strength development on shell-powder pastes and concretes at several replacement levels, particle sizes and organic contents; (ii) chloride migration/diffusion, acid-immersion, freeze–thaw, fatigue and abrasion tests on same-w/b and strength-matched mixes; (iii) further out-of-sample comparisons of predicted and measured properties on oyster shell and on durability indicators, beyond the single strength dataset used in Section 4.4; and (iv) a plant- and supply-chain-specific life-cycle inventory.

9. Conclusions

- A coupled hydration–transport–damage–carbon framework with paired shell/OPC comparison, Latin hypercube Monte Carlo, Sobol sensitivity and a falsification region was developed and verified numerically. Its inert-filler strength model was rejected by published scallop-shell data (predicted 0.43 against 0.75 relative strength at 40% replacement); a filler efficiency (posterior median 0.65) was calibrated on two mixes and tested on a third. Transport and durability predictions remain uncalibrated.
- At constant w/b, shell replacement of cement lowers embodied carbon roughly in proportion to r (−21.2% at 25%) but reduces strength to 66% of OPC and shortens median service life in the pavement and culvert cases; the 80% carbon claim is reached only when strength is lost.
- When strength is restored by adding binder, the carbon benefit disappears (−0.3% at 25%) while predicted life increases, but by an amount that depends on the diffusivity model; a joint advantage of

at least 5% carbon reduction with non-inferior life has a probability of at most 0.01 for P1 at r = 25% in the original model.

- With the calibrated filler efficiency the pavement joint-advantage probability rises to 0.23–0.28 at r = 25% (carbon change −21% same w/b, −16% strength-matched) while the marine culvert stays at 0.00: the claim is conditional on the strength model and on an untested chloride transport mapping, not established.
- Shell calcium carbonate used as limestone feedstock (P2) does not reduce carbon unless calcination CO_2 is credited as biogenic; durability is essentially unchanged.
- Carbon claims for shell concrete are controlled by supply-chain logistics, not by material chemistry, and durability claims by the filler efficiency and the diffusivity of filler-bearing pastes; these are the quantities to measure first.
- Shell used in concrete is not available as reef substrate; the allocation of recovered shell between the two uses is treated in a companion paper (Sandler 2026b).

Declarations

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CRediT authorship contribution statement. Leon Sandler: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the author used Claude (Anthropic) to help develop and test the simulation code, run the analyses and draft and revise the text and figures. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Data availability. All code, automated checks, result files and manuscript sources are openly available at https://github.com/sandlerleon/oyster-shell-concrete-mc (release v1.0.0; MIT licence) and archived on Zenodo (software, https://doi.org/10.5281/zenodo.23867048; preprint of this manuscript, https://doi.org/10.5281/zenodo.23867044). The published strengths used for calibration are from Abbas and Kudukkan (2025) and are tabulated in Table 5. No new experimental data were collected. The companion allocation study (Sandler 2026b) is archived at https://doi.org/10.5281/zenodo.23867050.

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Appendix A. Model parameters

All ranges are literature-style placeholders. tri(lo, mode, hi); uni(lo, hi); logn(median, σ in ln units); logu(lo, hi) is log-uniform. Parameters in the upper block are common to all cases; case-specific parameters follow.

Parameter	Key	Distribution	Unit
Hydration rate	k0	logn(0.26, 0.2)	1/d
Reaction order	nh	uni(1.5, 2.5)	-
Hydration Ea	Ea_h	uni(3.0e+4, 4.5e+4)	J/mol
Max. hydration	acap	uni(0.9, 0.97)	-
Gel strength S0	S0	logn(78, 0.1)	MPa
Strength exponent	kap_s	uni(1.8, 3)	-
Filler nucleation	nu	uni(0, 1.5)	-
Packing bonus	bpack	uni(0, 0.3)	-
Shell organics	o_org	tri(5.0e-3, 0.02, 0.04)	kg/kg
Washing efficiency	eta_w	uni(0.3, 0.9)	-
Organic retardation	kap_o	logu(5, 200)	-
Impurity penalty	kimp	uni(0, 0.05)	-
D28 factor	lamD	logn(1, 0.4)	-
D28 model form	chi_D	uni(0, 1)	-
Porosity exponent	dzeta	uni(-0.3, 0.3)	-
Age exponent m	m0	tri(0.12, 0.2, 0.3)	-
m shift	omega	uni(0, 0.3)	-
Crack–diffusion coupling	kapc	uni(2, 20)	-
Diffusion Ea	UD	uni(3.5e+4, 5.0e+4)	J/mol
Critical chloride	Ccrit	tri(0.6, 1, 1.6)	kg/m3
Propagation period	tprop	tri(3, 6, 12)	yr
Shrinkage strain	eps0	tri(400, 600, 900)	microstrain
Shrinkage time	tau_sh	tri(20, 35, 60)	d
Restraint	KR	uni(0.2, 0.6)	-
Creep relaxation	psi	uni(0.4, 0.7)	-
Early crack damage	d_cr	uni(0.03, 0.1)	-
Curling factor	f_env	uni(0.05, 0.25)	-
Air content	a_air	tri(0.02, 0.05, 0.07)	m3/m3
Critical saturation	S_crit	uni(0.8, 0.92)	-
FT damage/cycle	d1	logu(3.0e-4, 3.0e-3)	damage/cycle
FT exponent	q_ft	uni(1, 3)	-
Load variability	var_t	tri(0.85, 1, 1.15)	-
Traffic cycles/yr	n_traf	logu(5.0e+4, 1.0e+6)	cycles/yr
Failure damage	Dfail	tri(0.2, 0.3, 0.4)	-
Heat of hydration	Qcem	uni(350, 480)	kJ/kg
Thermal drop	psiT	uni(0.5, 0.9)	-
Thermal restraint	Rres	uni(0.3, 0.7)	-

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Parameter	Key	Distribution	Unit
Thermal expansion	alphaT	uni(8.0e-6, 1.2e-5)	1/K
Leaching saturation	Csat	uni(15, 22)	mol/m3
Leachable Ca	xi_L	uni(0.3, 0.7)	-
Permeability factor	kK	logu(0.03, 1)	1/m
Leaching limit	x_lim_L	tri(30, 50, 100)	mm
Ca diffusivity	Dratio_Ca	uni(0.5, 1.5)	-
Acid-accessible Ca	xiN	uni(0.4, 0.8)	-
Calcite access size	dc	logu(50, 300)	um
H+ diffusivity	DH	uni(3, 8)	-
Acid limit	x_lim_A	tri(10, 20, 30)	mm
Abrasion rate	k_ab	tri(0.05, 0.3, 1)	mm/yr
Abrasion exponent	b_ab	uni(1.5, 2.5)	-
Wear limit	w_lim	tri(15, 25, 40)	mm
Cement EF	EFcem	tri(0.75, 0.85, 0.95)	kgCO2e/kg
Raw-meal CaCO3	mCaCO3	tri(1, 1.1, 1.2)	kg/kg
Aggregate EF	EFagg	tri(4.0e-3, 7.0e-3, 0.012)	kgCO2e/kg
Water/admixture EF	EFmisc	tri(5, 10, 20)	kgCO2e/m3
Shell drying energy	e_dry	tri(20, 60, 150)	kWh/t
Grinding energy	E_ref	tri(25, 45, 90)	kWh/t
Crushing energy	E_cr	tri(5, 10, 20)	kWh/t
Electricity EF	EF_el	tri(0.1, 0.4, 0.8)	kgCO2e/kWh
Heat EF	EF_th	tri(0.2, 0.23, 0.3)	kgCO2e/kWh
Shell haul distance	dist_sh	tri(30, 150, 600)	km
Truck EF	EF_tr	tri(0.07, 0.1, 0.15)	kgCO2e/t-km
Landfill credit	c_land	tri(0, 5.0e-3, 0.02)	kgCO2e/kg
Limestone EF	EF_lq	tri(3.0e-3, 6.0e-3, 0.012)	kgCO2e/kg
Kiln penalty	kiln_pen	tri(0, 5.0e-3, 0.02)	kgCO2e/kg
Shell purity	p_sh	uni(0.92, 0.98)	-

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Case-specific parameters. Case A (pavement) / Case B (dam) / Case C (culvert):

Parameter	A	B	C	Unit
Binder content	tri(320, 350, 400)	tri(180, 230, 280)	tri(340, 370, 420)	kg/m3
w/b	tri(0.38, 0.42, 0.46)	tri(0.45, 0.5, 0.6)	tri(0.38, 0.42, 0.46)	-
Surface chloride	tri(2, 5, 9)	tri(0, 1.0e-4, 2.0e-4)	tri(3, 6, 12)	kg/m3
Cover	tri(50, 70, 100)	tri(50, 70, 100)	tri(50, 65, 90)	mm
Service temperature	tri(2, 10, 18)	tri(4, 10, 16)	tri(8, 16, 26)	C
Relative humidity	uni(55, 90)	uni(90, 100)	uni(85, 100)	%
Air-void filling	uni(0.1, 0.5)	uni(0.2, 0.7)	uni(0.3, 0.9)	-
Freeze–thaw cycles/yr	tri(10, 40, 90)	tri(0, 15, 50)	tri(0, 20, 60)	cycles/yr
Curing temperature	tri(5, 20, 35)	tri(10, 15, 25)	tri(5, 20, 35)	C
Construction emissions	tri(15, 30, 55)	tri(10, 20, 40)	tri(10, 25, 45)	kgCO2e/m3
Hydraulic gradient	–	tri(5, 20, 60)	–	-

Filler efficiency (extension, Section 4.4). $k_f \sim U(0, 1.2)$; $k_f = 0$ reproduces the original model; calibrated by rejection ABC on S0F15 and S30F15 (tolerance 0.07).

Scenario variables: stress-ratio classes (low, medium, high) = 0.28, 0.38, 0.46 of the OPC flexural strength; acid concentration (mild, moderate, severe) = 0.03, 0.3, 3 mol H^+/m^3; chloride multiplier = 0.5, 1, 2; leaching-saturation multiplier = 0.5, 1, 1.5. Other fixed values: R = 0.1 in the S–N relation, horizon 100 yr, time step 0.125 yr, $D_f \sim$ tri(0.2, 0.3, 0.4) (a parameter), shell and cement densities 2710 and 3150 kg/m^3.

Appendix B. Symbols and units

Symbol	Meaning	Unit
α, α_{max}	Hydration degree; attainable degree	–
k, k_0	Hydration rate constant (at 20 °C)	1/d
r, ρ	Shell replacement of cement (P1); of limestone (P2)	–
v_c, v_f, v_w	Volumes of cement, shell filler and water per m^3 of concrete	m^3/m^3
φ_{cap}, φ_p	Capillary porosity (per concrete; per paste)	m^3/m^3
X	Gel/space ratio	–
f_c, f_t, f_r, E	Compressive, tensile, flexural strength; modulus	MPa
D_{28}, D(t)	Chloride diffusivity at 28 d; at age t	m^2/s
C_s, C_{crit}	Surface and critical chloride content	kg/m^3
τ	Time integral of diffusivity	m^2
x, N, C_a	Front depth; capacity per volume; aggressive concentration	mm; mol/m^3; mol/m^3
D, D_m, D_f	Damage, mechanical damage, failure damage	–
S, S_{eff}, S_w	Stress ratio, effective stress ratio, degree of saturation	–
N_f	Cycles to failure	cycles
n_{traf}, n_{ft}	Traffic cycles; freeze–thaw cycles per year	1/yr
d_{ft}	Freeze–thaw damage per cycle	–
L	Service life (censored at 100 yr)	yr
E_{gate}, E_{total}	Cradle-to-gate; total embodied carbon	kg CO_2e/m^3

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Symbol	Meaning	Unit
β_bio	Credited fraction of calcination CO_2	–
$\mathscr{F}$	Falsification region	–

Appendix C. Reproducing the results

```
pip install numpy scipy matplotlib
python ShellConcreteMC_v2.py --verify
python ShellConcreteMC_v2.py --reproduce --n 3000 --sobol-n 1024 --outdir out
python calibrate_abbas.py          # out-of-sample test and ABC calibration (ShellConcreteMC_v3.py)
python kf0_prediction.py           # original-model predictions for the published mixes
python make_fig_calibration.py out
```

Studies use seed 21, severity/load scans seed 31, Sobol seed 11; the figures use fixed seeds in the script. A single full run is long on a single core; the results here were generated in stages (study, Sobol per case, figures), which the script supports through its individual functions.

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